

PROJECT PROPOSAL

MediaPipe Interactive Web Site

Air Canvas · Gesture Game · Gesture Quiz

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1. PROJECT OVERVIEW

Project Topic

The MediaPipe Interactive Web Site is a multi-mode, browser-based AI application that leverages real-time hand gesture recognition to provide three distinct interactive experiences: an Air Drawing Canvas, a Gesture-Controlled Snake Game, and a Hand Gesture Quiz. Users are presented with a unified home screen from which they select any mode. A single MediaPipe hand-tracking pipeline powers all three modes, with a Flask backend handling AI processing and an HTML/CSS/JavaScript frontend delivering the interface.

Objective

The primary goal is to develop a fully integrated, gesture-driven web application that demonstrates how computer vision can replace traditional input devices to create accessible and innovative user experiences. The system aims to:

- Enable users to draw on a digital canvas using only their index finger in the air
- Control a browser-based Snake game using hand tilt and gesture directions
- Answer multiple-choice quiz questions by holding up a number of fingers
- Deliver all three experiences through a single cohesive website with no paid APIs

2. SYSTEM DESCRIPTION

Original Problem Background

Traditional computer interaction relies entirely on physical input devices such as keyboards, mice, and touchscreens. This creates barriers for users with motor impairments, limits creative expression, and reduces engagement in educational or gaming contexts. There is a growing need for natural, contactless human-computer interaction that runs directly in a web browser without specialized hardware. Existing gesture-based solutions are either too expensive, require dedicated devices, or are not accessible to general users.

Innovations Introduced

- Real-Time Gesture Recognition via MediaPipe: The system uses Google's MediaPipe Hand Landmarker to detect 21 hand keypoints per hand at real-time speed through a standard webcam, requiring no special hardware or paid services.
- Multi-Mode Architecture: A single Flask application manages a unified MediaPipe pipeline and routes gesture data to three completely different interaction modes, all accessible from one home screen.
- Air Canvas — Contactless Drawing: Index fingertip coordinates detected by MediaPipe are streamed to the browser and used to draw on an HTML5 Canvas element in real time.
- Gesture Game — Hands-Free Gaming: Hand orientation and gesture classification control a Snake game rendered in the browser, with Python handling all gesture logic and sending commands via Flask routes.
- Gesture Quiz — Finger-Count Answering: A quiz system where users answer MCQ questions by raising 1, 2, 3, or 4 fingers. MediaPipe counts extended fingers and Flask evaluates the answer.

3. AI APPROACH AND METHODOLOGY

Search Algorithms

State-space search and rule-based decision logic are applied in multiple components:

- Finger State Classification: For each detected hand landmark, the system evaluates whether each finger is extended or folded by comparing joint positions — a constraint-based search across 5 binary states (one per finger).
- Gesture-to-Command Mapping: The quiz mode uses a lookup/matching algorithm to map finger count (1–4) to answer options (A, B, C, D), and the game mode uses directional state matching to determine movement commands.
- Snake Game Pathfinding: The game includes a simple BFS component that detects dead-end moves to prevent immediate wall collisions.

Computer Vision — MediaPipe

- MediaPipe Hand Landmarker model detects 21 3D hand landmarks per frame
- Custom Python logic calculates finger extension states, fingertip coordinates, and hand orientation angles
- OpenCV (cv2) captures webcam frames and annotates them before streaming to the browser

Automation & Backend

- Flask continuously runs the MediaPipe pipeline in a background thread and exposes REST endpoints (/video_feed, /get_gesture, /get_coords, /get_quiz_state) that the browser polls automatically
- JavaScript fetch() calls poll Flask every 100ms to receive live gesture data, making the system fully automated once started

4. SYSTEM RULES AND MECHANICS

System Workflow

User opens website → Selects mode from Home Screen → Webcam activates → MediaPipe begins processing → Gesture data streams to browser → Mode-specific interaction begins

Mode 1 — Air Canvas

- Index finger raised, others folded = Drawing mode active
- All fingers open = Pause / eraser
- Fist = Clear canvas
- Fingertip (x, y) coordinates sent via /get_coords to JavaScript
- JavaScript draws connecting lines on the HTML5 Canvas element in real time

Mode 2 — Gesture Snake Game

- Hand tilted left / right / up / down = Snake changes direction
- Fist gesture = Pause game
- Peace sign (two fingers) = Restart after game over
- Flask /get_gesture endpoint returns current direction string
- JavaScript Snake game reads direction every 100ms and moves accordingly

Mode 3 — Gesture Quiz

- Flask loads quiz questions from a Python list
- Question and 4 options (A/B/C/D) sent to browser via /get_question
- User holds up 1, 2, 3, or 4 fingers to select answer A, B, C, or D
- MediaPipe counts extended fingers; Flask checks against correct answer
- Score updates live on the page; final summary shown at end of quiz

Output / Success Conditions

- Air Canvas: Smooth continuous lines drawn accurately tracking fingertip position
- Gesture Game: Snake moves correctly in response to hand direction with no input lag
- Gesture Quiz: Finger counts detected reliably; score tracked and displayed accurately

5. LIBRARIES AND TOOLS

CATEGORY	TOOL / LIBRARY	PURPOSE
Computer Vision	MediaPipe (Hand Landmarker)	Real-time hand & finger detection
Computer Vision	OpenCV (cv2)	Webcam capture & frame annotation
Backend	Flask (Python)	Web server, REST API, video streaming
Frontend	HTML / CSS	UI structure and styling
Frontend	JavaScript	Game loop, canvas drawing, fetch polling
Python Libraries	NumPy	Landmark coordinate calculations
Python Libraries	Threading	Background MediaPipe processing
Rendering	HTML Canvas API	Snake game & air canvas drawing surface
Version Control	Git / GitHub	Project management

6. REFERENCES

- Google MediaPipe Documentation — Hand Landmarker Task Guide.
- Flask
- HTML Canvas API
- OpenCV Python Documentation — VideoCapture and Image Processing
- Python
- Russell, S. & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.